

Hafiz Muhammad Usama Saleem

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Portfolio: [Website](#) | [LinkedIn: Profile](#) | [GitHub: Profile](#)

Machine Learning Engineer specializing in LLMs, RAG systems, and voice automation. Experienced in building AI agents, fine-tuning models, and deploying scalable GenAI applications using LangChain, CrewAI, and FastAPI. Passionate about creating intelligent, cost-effective AI systems that drive measurable business impact.

EDUCATION

MSc in Data Science

FAST NUCES, 2025

Postgraduate Diploma in Data Science and Artificial Intelligence

NED University, 2024

Bachelor of Engineering in Electrical

NED University, 2020

WORK EXPERIENCE

AI Engineer at Intersys Ltd (August 2025 - Current)

- Built an AI-powered Voice Bot using Twilio, STT, TTS, and LLMs (OpenAI & open-source), integrated directly with the company's POS system.
- Enabled real-time order management customers could place, modify, and track orders via phone, with the bot updating the POS automatically.
- Delivered human-like conversations that replaced full time call agents, drastically improving customer experience at a fraction of the cost.
- Scaled on AWS to ensure 24/7 reliability, high availability, and production-grade performance.
- Drove significant cost savings by automating operations, eliminating the need for dedicated call staff, and streamlining order workflows.

Machine Learning Engineer at PixelEdge Solutions (Pvt.) Limited (October 2023 – August 2025)

Fintech Product (Data Automation & ML Integration)

- Built and deployed ML algorithms to automate data extraction workflows, significantly reducing manual effort and generating structured Excel reports from raw financial inputs.
- Applied advanced preprocessing, feature engineering, and exploratory data analysis on financial datasets to support KPI prediction and decision-making.

Sustainability / Environment-Focused Product (Intelligent Document Assistant)

- Developed a **RAG-based** conversational solution using **LangChain** to answer predefined user queries by retrieving insights from internal sustainability and policy documents..
- Integrated ChatGPT API to enhance natural language understanding, enabling accurate, human-like responses grounded in document context.
- Designed and deployed robust REST APIs using **Flask** and **FastAPI** to serve ML models and conversational agents across internal tools.

Agentic AI Systems (Cross-Domain Application)

- Built **multi-agent systems** using frameworks like **SmolAgent**, **CrewAI**, and **LangGraph** to automate complex research and data retrieval workflows.

- Developed a Gemini-powered agent to autonomously extract company insights using search-based browsing and reasoning over multi-source data.

Trainee Data Scientist at Techma Zone (January 2023-October2023)

- Performed data preprocessing, cleaning, and feature engineering on real-world datasets to prepare for analysis and modeling.
- Conducted exploratory data analysis (EDA) and visualized insights using tools like Matplotlib, Seaborn, and Plotly.
- Collaborated in cross-functional projects to apply machine learning algorithms and evaluate model performance using industry-standard metrics.

TECHNICAL SKILLS

- **GenAI:** LangChain, LLM Integration, Prompt Engineering, RAG, Vector Databases
- **Agentic AI Frameworks:** CrewAI, SmolAgent, LangGraph
- **Machine Learning:** Supervised & Unsupervised Learning, Transformers, Model Evaluation
- **Deep Learning:** PyTorch, TensorFlow, Keras
- **Natural Language Processing:** NLTK, spaCy, Text Classification, Summarization
- **Voice AI:** Whisper, Twilio, Speech-to-Text (STT), Text-to-Speech (TTS)
- **Programming Languages:** Python, SQL, JavaScript (basic understanding)
- **Databases:** PostgreSQL, MySQL
- **Cloud & DevOps:** AWS, Docker, GitHub Actions, CI/CD
- **Model Deployment & APIs:** Flask, FastAPI, REST APIs, Authentication & Authorization
- **Frameworks & Tools:** Scikit-learn, Hugging Face, Jupyter Notebook, Git, Pandas, NumPy
- **Data Engineering & Visualization:** ETL Pipelines, Semantic Chunking, Matplotlib, Seaborn, Plotly

PERSONAL PROJECTS

Mini AI Concierge (Guest Support Bot) [GitHub](#) | [Loom Demo](#)

Built an LLM-powered chatbot using RAG architecture to automate guest support for vacation rentals. Integrated document retrieval and fallback escalation logic using FastAPI, FAISS, and Gemini.

Agentic Travel Companion [GitHub](#) | [Loom Demo](#)

AI-powered Streamlit application that transforms travel planning into an interactive, personalized experience. Utilizes multiple specialized AI agents to assist with destination selection, budgeting, and trip itineraries using CrewAI.

Stock Price Prediction Model [GitHub](#)

Developed a logistic regression model to predict stock price movement based on financial indicators, using exploratory data analysis and feature selection techniques.

CERTIFICATION

Generative AI with LLMs | Coursera

Complete Machine Learning & Data Science Bootcamp | Udemy

100 Days of Code: The Complete Python Pro Bootcamp | Udemy

Mathematics for Machine Learning and Data Science Specialization | Coursera

CORE COMPETENCIES

Problem-Solving | Communication | Adaptability | Time Management

LANGUAGE SKILLS

English | Urdu